

CALRIG · DESIGN DECISION RECORD

CalRig recommendations accepted

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DOCUMENT CONTROL

Document ID	CLR-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/calrig		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 (collocation partner) decided by Amish on 2026-10-02
0.2	2026-09-26	Amish Chadha	Budget set to \$412 to cover the priced BOM: decided by Amish, 2026-09-26 (O2 closed)
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below marked "Decided by Amish, 2026-09-25: go with recommendation" is accepted; items without a recommendation remain open, except O2 (budget), decided by Amish on 2026-09-26, and O1 (collocation partner), decided by Amish on 2026-10-02.

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." This record lists every item in `docs/REVIEW.md` (TRL 2 and TRL 3 sessions) and in CLR-DDR-001 that carried a recommendation, now decided, and what changed in the repo. Items without a recommendation stay "Proposed, awaiting Amish". Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, and `trl` and `trl_target` stay at 3.

■ Options considered

The options for each item are those listed in `docs/REVIEW.md` (sessions 2026-09-25, /populate and TRL 3) and in CLR-DDR-001.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Budget (DDR-001 A1)	Raise <code>budget_usd</code> from \$300 to \$400	<code>project.yaml</code> <code>budget_usd</code> : 400 ; R12 target restated as \$400 in CLR-REQ-001 v0.4; README, CLR-PRB-001 v0.4, CLR-PRC-001 v0.4, BOM notes and CLR-CAL-001 v0.2 updated. The design is \$412, so R12 is still not met
D2	Reference strategy (DDR-001 A2)	Transfer references checked against salt fixed points, an ice point and field collocation; uncertainty chain in every report	Wording only: "decided" in CLR-PRC-001 v0.4 and CLR-DDR-001 v0.2
D3	Cooling limit (DDR-001 A3)	Accept 10 °C only in rooms at 25 °C or below; R1 as redefined in CLR-REQ-001 v0.3	Wording only
D4	Test aerosol (DDR-001 A4)	Incense smoke decay; aerosol type stated in every report	Wording only
D5	Gas scope (DDR-001 A5)	NO2 by field collocation only; align AirStreet's wording	CLR-PRB-001 v0.4 wording. AirStreet's pitch line is listed as a cross-repo action in <code>docs/REVIEW.md</code> ; AirStreet is not edited from this repo
D6	Door condensation (DDR-001 A6)	Insulated removable door panel	Wording only (already in the model and BOM)
D7	Inner Peltier sink (R2)	Larger inner fin block, about 0.20 K/W	<code>cad/src/model.py</code> : inner sink 30 x 80 x 90 mm to 45 x 120 x 110 mm and <code>sink_in_r</code> 0.20 K/W; STEP and STL re-exported; CLR-DWG-001 Rev P1 to P2; BOM line 5 \$30 to \$38; media regenerated. CLR-CAL-001 v0.2: inner sink at 20 °C, 85 % RH in a 22 °C room 16.2 °C to 18.3 °C (0.9 K above the dew point), dry up to about 28 °C (was 19 °C); R2 not met to met. Side effects: lowest chamber temperature in a 25 °C room 8.9 °C to 6.7 °C, in a 30 °C room 13.3 °C to 11.1 °C; cooling to 10 °C 3.7 h to 2.4 h; sweep plus particle run 5.6 h to 5.5 h; peak power 89 W to 90 W; mass 13.4 kg to 13.6 kg
D8	Mass (R13)	Relax R13 from 12 kg to 14 kg; it is a bench rig	CLR-REQ-001 v0.4 R13 restated; CLR-CAL-001 v0.2 R13 not met to met (13.6 kg)
D9	Reference temperature (R5)	Add a certified thermistor probe if the budget allows	The budget does not allow it (\$412 against \$400), so no probe is added to the BOM; R5 stays at risk. Recorded in CLR-PRC-001 v0.4, CLR-REQ-001 v0.4 and CLR-CAL-001 v0.2. Buying the probe would also be TRL 4 purchasing, which is on hold

Table 2. Items still open.

#	Item	Status
O1	Collocation partner for the transfer SPS30. No recommendation was made.	Decided by Amish, 2026-10-02: a state or local regulatory monitoring site with a federal equivalent PM2.5 monitor; first candidate to approach a Texas Commission on Environmental Quality site in Dallas-Fort Worth, with South Coast AQMD's AQ-SPEC program as the fallback (CLR-DEC-001)
O2	Budget figure above \$400. The design is \$412; options are to accept about \$412, drop CO2 for a core version at about \$345, or find \$12 of savings. No recommendation was made.	Decided by Amish, 2026-09-26: budget set to \$412 (see below)

Budget approved, 2026-09-26

On 2026-09-26 Amish wrote: "i approve all the budget items."

- Budget set to \$412 to cover the priced BOM: decided by Amish, 2026-09-26. This closes O2. `project.yaml` `budget_usd` 400 to 412; CLR-REQ-001 R12 target \$412, status Not met to Met; CLR-CAL-001 v0.3 (`sizing.py` rerun); CLR-PRB-001, CLR-PRC-001, README, `bom/bom-notes.md` and the blueprint key figures updated. The BOM and geometry are unchanged. D9 (certified probe if the budget allows) is unchanged: the new budget has no headroom for it.

■ Consequences

- Requirement status (CLR-CAL-001 v0.2): met 12 (was 10), at risk 3, not met 2 (was 4), not verifiable at TRL 3 1. Not met: R8 (collocation site, O1) and R12 (\$412 against \$400, O2).
- After the 2026-09-26 budget approval (CLR-CAL-001 v0.3): met 13, at risk 3, not met 1 (R8), not verifiable at TRL 3 1.
- Documents bumped: CLR-PRB-001 v0.4, CLR-PRC-001 v0.4, CLR-REQ-001 v0.4, CLR-CAL-001 v0.2, CLR-DDR-001 v0.2; drawing CLR-DWG-001 Rev P2.
- Cross-repo action: AirStreet to align its "calibrated on CalRig" pitch line with NO2 by field collocation (D5). Not edited here.
- TRL 4 work (a built chamber, lab tests, buying parts, firmware beyond a sketch) remains on hold.